

Islandwide Cooperative Retail and Distribution System

Requirements Clarification Questionnaire

A structured set of questions to be answered by the federation before solution design begins. Each question states plainly why it is being asked and what part of the system it determines. Abbreviations are expanded in the glossary at Section 4.

Document	Requirements clarification questionnaire
Status	Draft for discussion
Prepared for	Central cooperative and member cooperative societies
Scope	Central supply, shop requisition and fulfilment, multi-location inventory, point of sale, inter-entity settlement, reporting
Structure	20 question sections, approximately 150 questions
Next step	Answer the eight gating questions in Section 3 first; the remainder depend on them

How to read this document

This document deliberately asks more than it answers. In a federation of independent societies, most of the difficult decisions are commercial and governance decisions rather than technical ones. Recording them explicitly now is far cheaper than discovering the disagreement during build or, worse, after go-live.

1. Purpose and how to use this document

The proposed system serves a federation, not a chain. The central cooperative and each member cooperative shop are separate legal entities. That single fact changes the nature of almost every transaction in the system: goods moving from central to a shop are not an internal stock transfer but a commercial transaction between two businesses, and data generated at a shop is that shop's own commercial record rather than central's.

The questions below are grouped by domain. Within each domain, questions are numbered with a section letter and sequence number so they can be referenced unambiguously in discussion and in the answers you return.

Conventions used

Element	Meaning
A-01	Question reference. Section letter plus sequence number. Please answer against these references.
Why we ask	Included where the reason for the question is not self-evident, or where the answer has consequences beyond the obvious.
Shaded box	Context, a decision already taken, or a warning about a common failure mode.

Who needs to answer

These questions are not all for one person. Broadly: Sections A, C, G, K and S need the federation board or its equivalent decision-making authority. Sections I, J and L need the finance function at central and a representative shop treasurer. Sections B, D, E and F need operations and warehouse management. Sections H, M, N, P, Q and R need whoever will own the system after go-live. Where a question cannot be answered by any single party, that is itself a finding worth recording.

2. Points already settled

The following have been confirmed in discussion and are treated as fixed assumptions in the questions that follow. If any of these change, the dependent questions are flagged in the relevant section.

Point	Position	Consequence
Existing systems	There is no ERP (Enterprise Resource Planning system) anywhere in the federation. Some shops may run a point of sale (POS) system.	No ERP migration. The new system supplies POS, so any existing POS is replaced rather than integrated. Data extraction from those systems is still an open question (Section R).
Connectivity	Internet access will be provided to all shops, but outages and slow links are expected in remote areas.	Offline capability at the till is a hard requirement, not an option. How far offline capability must extend is the subject of Section H.
Languages	Sinhala, Tamil and English must all be supported.	Affects interface, master data, printed output, search and sorting, and receipt printer selection. Depth of the requirement is explored in Section P.
Stock losses	Damages and discards must be handled by the system.	Treated as a full domain in Section F, and tightly coupled to the goods ownership decision in Section A.

3. The gating questions

Eight questions determine the shape of everything else. Until they are answered, detailed design cannot begin, because the answers change the data model, the ledger and the governance rules rather than merely adding features. We recommend these are settled first, in writing, by whoever holds decision-making authority for the federation.

Ref	Decision	What it determines
A-01	Do goods become the shop's property on receipt (firm sale), or does central retain ownership until sale (consignment)?	Whose balance sheet holds the stock, when central recognises revenue, who carries the loss on damage and expiry, and the entire credit exposure model.
B-01	At what level is a Stock Keeping Unit (SKU) defined?	Every stock figure, sales report and reorder calculation aggregates at this level. If central and shops draw the line differently, no report will reconcile.
B-02	Is a case or outer pack a separate SKU, or the same SKU with a pack conversion?	The single most common source of stock error in distribution systems. Central thinks in cases; the counter sells singles.
M-01	Who legally owns the transaction data generated at a shop?	What central may see and do with shop data, what happens when a shop leaves the federation, and who answers to a regulator.
J-01	Full accounting per shop, or only the receivable and payable between central and each shop?	One of the largest scope decisions in the programme. Full multi-entity accounting roughly doubles the finance scope.
H-01	What is the longest offline period the till must survive and still trade?	An hour is a simple queue. Several days is a full local database with conflict resolution. The cost difference is substantial.
K-01	Do members need to be distinguished from ordinary customers at all?	If no, most of the membership domain disappears. If yes, member identification enters the checkout flow and affects queue speed.
S-03	Is a phased first release acceptable, or must full scope go live at once?	Usually the difference between a programme that lands and one that stalls.

4. Glossary of abbreviations and terms

Terms are expanded here and used in abbreviated form thereafter. Where a term has a specific meaning in this context, that meaning is given rather than the dictionary definition.

Term	Expansion	Meaning in this context
3PL	Third-Party Logistics	An external contractor performing warehousing or delivery on behalf of central.
AGM	Annual General Meeting	The members' meeting at which a cooperative's governance decisions are ratified.
API	Application Programming Interface	A defined way for another system to exchange data with this one automatically.
AP	Accounts Payable	Money an entity owes to others. From a shop's point of view, what it owes central for goods supplied.
AR	Accounts Receivable	Money owed to an entity. From central's point of view, what the shops owe it. The same transaction is AR for central and AP for the shop, which is why the two ledgers must agree.
COGS	Cost of Goods Sold	The cost of the stock that was sold, as opposed to the price it sold for. The difference is gross margin.
DO	Delivery Order	The document accompanying goods from central to a shop, listing what was dispatched.
EAN / GTIN	European Article Number / Global Trade Item Number	The standard barcode number printed on retail packaging.
ERP	Enterprise Resource Planning	An integrated business system covering finance, inventory and operations. Confirmed as not present today.
FEFO	First Expired, First Out	A picking rule that issues the stock nearest its expiry date first. Used where goods perish.
FIFO	First In, First Out	A picking or costing rule that treats the oldest stock as issued first.
FMCG	Fast-Moving Consumer Goods	Everyday grocery and household items sold in high volume at low margin.
GL	General Ledger	The complete set of accounting records for a legal entity.
GRN	Goods Received Note	The record a shop raises when it receives and counts a delivery. The point at which the shop's stock records are updated.
KPI	Key Performance Indicator	An agreed measure used to judge performance, for example shrinkage percentage.
MDM	Master Data Management	The discipline of governing shared reference data such as the product catalogue.
MFA	Multi-Factor Authentication	Requiring a second proof of identity beyond a password.
MOQ	Minimum Order Quantity	The smallest quantity that may be ordered for a given item.

Term	Expansion	Meaning in this context
PII	Personally Identifiable Information	Data identifying an individual, such as a member's name, address or identity card number. Subject to privacy obligations.
PLU	Price Look-Up code	A short numeric code keyed at the till for items without a barcode, typically loose produce.
PO	Purchase Order	An order placed by central on an external supplier.
POS	Point of Sale	The till system at the shop counter where sales to customers are recorded.
RBAC	Role-Based Access Control	Granting permissions by job role rather than to individuals one by one.
RPO	Recovery Point Objective	The maximum amount of data, expressed as a period of time, that the business is willing to lose in a disaster. Drives backup frequency.
RTO	Recovery Time Objective	The maximum time the business is willing to be without the system in a disaster. Drives failover design and cost.
SKU	Stock Keeping Unit	The smallest distinctly tracked and counted item in the catalogue. Its definition is a business decision, addressed in Section B.
SLA	Service Level Agreement	A commitment on availability and support response, with consequences for breach.
SSO	Single Sign-On	One login granting access to several systems.
STO	Stock Transfer Order	An instruction to move stock between locations. Note that between independent cooperatives this is a sale, not a transfer.
SVAT	Simplified Value Added Tax	A credit voucher scheme operating alongside VAT in certain supply chains.
UOM	Unit of Measure	The unit in which an item is counted, bought or sold, for example kilogram, each, or case of twenty-four.
VAT	Value Added Tax	Consumption tax charged on sales and reclaimable on purchases by registered entities.
Consignment		An arrangement in which the supplier retains ownership of goods until the retailer sells them, and is paid only on sale.
Firm sale		An arrangement in which ownership and risk pass to the buyer on delivery, regardless of whether the goods are subsequently sold.
Shrinkage		The gap between stock the records say should be present and stock actually counted, after known losses are accounted for.

Section A. Commercial and ownership model

Why this section comes first

This is the most consequential section in the document. Because central and each shop are separate legal entities, goods moving between them is a real commercial transaction. Whether ownership passes on delivery or is retained until sale determines the inventory model, the accounting entries, the credit risk, and who absorbs losses. Nearly every later section inherits from A-01.

- A-01** When central supplies goods to a shop, does ownership pass to the shop at that moment (firm sale), or does central retain ownership until the shop sells the item to an end customer (consignment)?

Why we ask: This determines whose balance sheet the stock sits on, when central recognises revenue, whether central can recall unsold stock, and who carries the loss when goods expire or are damaged on the shelf. The inventory model, the ledger postings and the credit exposure all follow from this one answer.

- A-02** Could the model differ by product category, for example firm sale for dry goods but consignment for high-value or short-shelf-life items?

Why we ask: If the model is mixed, every stock lot must carry an ownership flag rather than the system applying one global rule. That is considerably more complex and is far better known now than discovered during build.

- A-03** At what precise point does risk of loss transfer: when goods leave the central warehouse, when the vehicle arrives at the shop, or when the shop signs the GRN?

Why we ask: Determines who absorbs in-transit damage and theft, and therefore which party the system charges when a delivery arrives short or damaged.

- A-04** Is a shop obliged to accept and pay for stock that central allocates to it, or may it decline a delivery or return unsold stock?

Why we ask: Governs whether a returns-to-central workflow and a credit note mechanism are needed, and whether central can push stock or only respond to pull.

- A-05** If returns to central are permitted, under what conditions: time window, condition of goods, restocking charge, prior authorisation?

- A-06** At what price do shops buy from central: a wholesale or transfer price, cost plus a fixed margin, or a member-discounted rate?

- A-07** Is there a single price list for all shops, or do transfer prices vary by shop, region, or volume purchased?

Why we ask: Volume-tiered or regional pricing requires a price-list engine with effective-dated versions rather than a single price field on the product record.

- A-08** May shops purchase the same goods from outside suppliers if the price is better, or must they source exclusively from central where central stocks the item?

Why we ask: Determines whether the system must record a parallel non-central supply chain per shop, and whether central can treat shop requisitions as a reliable proxy for true demand when planning procurement.

- A-09** What are the payment terms: does the shop pay before dispatch, on delivery, or on credit? If on credit, over what period?

- A-10** Is there a credit limit per shop, and should the system block a requisition that would breach it, or merely warn?

Why we ask: A hard block has real operational consequences, since a shop that cannot order cannot trade. The rule and its escalation path must be agreed by the federation before it is built, not chosen by a developer.

- A-11** Who may override a credit block, and must that approval be recorded for audit?

- A-12** Are penalties or interest charged on overdue balances, and must the system calculate them?
- A-13** Is there any obligation on central to supply, or on a shop to buy a minimum volume, that the system should monitor and report against?

Section B. SKU definition and product master data

On the SKU question

You asked what the scope of an SKU should be. This is the right question, and it is a decision rather than a lookup. An SKU (Stock Keeping Unit) is the smallest thing the business agrees to count as distinct. Every stock balance, sales report, reorder point and reconciliation aggregates at this level, so if central and the shops draw the boundary differently, no report between them will ever agree.

- B-01** At what level is a distinct SKU defined? Please confirm the exact attribute set that makes two items different SKUs: brand, product, pack size, variant, flavour, colour, grade?

Why we ask: This is the foundation of the entire data model. It should be written down as a rule that a warehouse clerk and a shop assistant would both apply the same way.

- B-02** Is a case, carton or outer pack a separate SKU from the single selling unit, or the same SKU with a defined pack conversion factor?

Why we ask: The most common source of stock error in distribution systems. Central habitually thinks in cases while the shop counter sells singles. Whichever answer is chosen, the conversion must be held in one place and be unambiguous.

- B-03** If the same product is bought from two different suppliers or manufacturers, is that one SKU or two?

Why we ask: One SKU simplifies selling and shelf management but hides supplier-level cost and quality differences. Two SKUs preserve traceability but fragment stock balances and sales reporting.

- B-04** Are batch number, manufacture date and expiry date attributes of the stock holding rather than of the SKU itself? Which product categories genuinely require batch or expiry tracking?

Why we ask: Batch tracking multiplies the size of the inventory records and changes the picking rule from FIFO to FEFO. It should be applied only where it is needed, because it adds counting effort at every shop.

- B-05** Are any items serialised, that is tracked by an individual unique number, for example appliances carrying a warranty?

- B-06** Are loose or weighed goods sold, such as rice, dhal, sugar or vegetables broken down from bulk? If so, how is the bulk purchase unit converted to a retail selling unit, and how is natural weight loss or gain accounted for?

Why we ask: Bulk-to-loose conversion creates a permanent and expected variance that is neither damage nor theft. Without a distinct transaction type for it, stock will never balance and shrinkage figures will be meaningless.

- B-07** Who has authority to create a new SKU in the central catalogue, and what approval does it require?

- B-08** May a shop create its own SKU for a locally sourced item that central does not supply?

- B-09** If shops may create local SKUs, should these sit in a separate numbering range from central SKUs, and should central be able to see them?

Why we ask: A separate range keeps the shared catalogue clean and prevents two shops inventing conflicting codes for the same product. Whether central can see them at all is a visibility decision that belongs with Section M.

- B-10** If a locally sourced item is later adopted into the central range, is there a process to promote or merge the local SKU into the central catalogue without losing that shop's sales history?

Why we ask: Merging product codes retrospectively is difficult if not designed for. Deciding now avoids a permanent break in trend data later.

- B-11** Do products carry barcodes reliably? What is done for unbarcoded and loose items: printed shelf labels, in-store printed barcodes, or PLU codes keyed at the till?
Why we ask: This determines the scanning workflow and the hardware specification at every shop, including whether label printers are needed.
- B-12** Can one SKU carry multiple barcodes, for example when a supplier changes pack artwork or an item is sourced from more than one market?
- B-13** What set of units of measure (UOM) is required, and who defines conversion factors: central only, or may a shop define its own?
- B-14** Must product names, descriptions and categories be held in all three languages in the master catalogue, and who is responsible for supplying and maintaining the translations?
Why we ask: Trilingual master data is a continuing content operation rather than a one-time technical feature. For a catalogue of any size it needs a named owner and a budget, otherwise it degrades into partial coverage within months.
- B-15** What product category hierarchy is required, and is it common across the federation, or may shops re-categorise items for their own merchandising?
Why we ask: A shared hierarchy is what makes cross-shop comparison possible. Local re-categorisation makes shops happier but breaks federation-level analysis unless a mapping is maintained.
- B-16** Are there items that are restricted or age-controlled, or that some shops are not permitted to sell?
- B-17** How are seasonal or discontinued items handled, and can an SKU be deactivated centrally while a shop still holds stock of it?
Why we ask: Deactivation while stock remains on shelves is a routine cause of tills refusing to sell saleable goods. The rule needs to be explicit.

Section C. Requisition, allocation and fulfilment

C-01 What triggers a requisition: manual entry by shop staff, a system suggestion based on reorder point and sales history, or both?

C-02 Does central approve every requisition, or is it fulfilled automatically when stock is available? Who approves, and are there value thresholds requiring higher approval?

C-03 When total requested quantity exceeds available central stock, how is the shortfall allocated across shops: first come first served, pro rata to request, share of historical sales, or a fixed quota?

Why we ask: This is a fairness question between independent member societies, with real commercial consequences for the shops that lose out. It should be decided by the federation and recorded in policy, not settled by whichever rule is easiest to code.

C-04 Should the allocation rule be configurable by central at any time, or fixed and changeable only by formal decision?

C-05 Is partial fulfilment permitted, and does the unfulfilled remainder stay open as a backorder or get cancelled?

C-06 Is there a minimum order quantity or minimum order value per requisition, per line, or per delivery run?

C-07 Are requisitions raised on a fixed cycle, for example each shop ordering on an assigned weekday, or ad hoc at any time?

Why we ask: A fixed cycle lets central batch picking and plan delivery routes efficiently. Ad hoc ordering demands continuous fulfilment capacity. This shapes the warehouse operation as much as the software.

C-08 May a shop amend or cancel a requisition after submission, and up to what point in the fulfilment process?

C-09 Are emergency or out-of-cycle requests permitted, and do they follow a different approval path or attract a different delivery charge?

C-10 May shops transfer stock directly to one another, or must all movement route through central?

Why we ask: A lateral transfer between two independent cooperatives is commercially a sale between two entities, requiring its own invoice and settlement. It also requires each shop to see something of another shop's stock position, which is a visibility decision.

C-11 If lateral transfers are permitted, does central approve them, and at what price do they transact?

C-12 Should a shop be able to see central stock availability before raising a requisition, and should that be a live figure or an indicative one?

Why we ask: A live figure is more useful but invites shops to race each other for scarce stock, which interacts with the allocation rule in C-03.

C-13 What lead time does central commit to between requisition and delivery, and should the system measure performance against it?

Section D. Delivery, goods receipt and discrepancies

D-01 Who performs delivery: a central fleet, a contracted third-party logistics provider, or does the shop collect from a depot?

D-02 Are deliveries planned as routes serving several shops from one vehicle?

Why we ask: If so, the system needs load planning and a per-drop delivery note rather than a simple per-shop dispatch document.

D-03 Should drivers use a mobile application for proof of delivery, or is a signed paper delivery order sufficient?

D-04 At goods receipt, does the shop count every line, or accept on trust and count later?

D-05 When the received quantity differs from the dispatched quantity, what happens? Does the shop record what it actually received and raise a discrepancy for central to investigate, or must the receipt match the dispatch note?

Why we ask: This decides whether a shop's stock ledger may temporarily diverge from central's records. Forcing an exact match makes reconciliation trivial but causes shops to carry stock on the system that they do not physically have, which corrupts every subsequent count.

D-06 Who investigates a disputed discrepancy, who adjudicates it, and within what time limit must it be raised?

D-07 When goods arrive damaged, is the shop credited automatically on receipt, or only after inspection and approval by central?

D-08 Should the receiving clerk be able to photograph damaged or short deliveries and attach the images to the discrepancy record?

Why we ask: Photographic evidence captured at the moment of receipt substantially reduces disputes between independent entities, and is inexpensive if designed in from the start rather than added later.

D-09 Should vehicle, driver and delivery timestamp be recorded for accountability?

D-10 Are returnable containers, crates or pallets in circulation that must be tracked and reconciled?

Why we ask: Unreturned containers are a genuine and often unmeasured cost. If they are in use, they need their own balance per shop.

Section E. Inventory management

E-01 How many central warehouses or regional depots exist, and does stock move between them?

E-02 Within a shop, must stock be distinguished by location, for example sales floor versus back store?

Why we ask: This is what lets a shop answer the question 'we have none on the shelf, is there any in the back'. It also introduces replenishment tasks, so it should be adopted deliberately rather than by default.

E-03 What costing method should apply: weighted average, FIFO, or standard cost?

Why we ask: This determines reported cost of goods sold and therefore each shop's declared profit. It has tax consequences and should be confirmed with the accountants who sign off the societies' books.

E-04 For batch-tracked goods, is the picking rule FIFO or FEFO?

E-05 How often are physical stock counts performed? Is cycle counting, that is counting a rotating subset continuously, acceptable, or are only full stocktakes recognised?

E-06 Must trading stop during a stocktake, or should the system support counting while the shop continues to trade?

Why we ask: Counting while trading is significantly more complex but avoids closing shops, which for a small society may mean a day of lost revenue.

E-07 Who may approve a stock adjustment arising from a count, and above what variance value must it be escalated?

E-08 Should reorder points and safety stock be calculated by the system for each shop and SKU, or entered manually by the shop?

E-09 Should the system permit negative stock, that is selling an item the records say is not present, or block the sale?

Why we ask: Blocking enforces data discipline but will stop genuine sales at the counter whenever records are wrong, which in early operation is often. Most retailers permit it and manage it through an exception report. This interacts directly with offline selling in Section H.

E-10 Are there items with special storage requirements, such as cold chain, that the system should track or alert on?

E-11 Should stock ageing be reported, showing how long stock has been held?

Why we ask: Ageing is the earliest warning of over-ordering, and under a consignment model it is central's exposure rather than the shop's.

Section F. Damages, discards, expiry and shrinkage

Coupled to the ownership decision

You raised this and it deserves treatment as a full domain. It is tightly coupled to Section A: under consignment, central absorbs most of these losses; under firm sale, the shop does. Until A-01 is answered, F-02 cannot be.

- F-01** Which loss categories must be recorded and reported separately? For example: damaged in transit, damaged in store, expired, spoiled, pest or weather damage, theft, unexplained shrinkage, staff consumption, samples and tasting, donation or charitable write-off, and returns to supplier.
Why we ask: Each has a different owner, a different accounting treatment and a different management response. Recording them all as a single 'adjustment' makes the data worthless for deciding what to fix.
- F-02** Under the ownership model chosen in A-01, who bears the cost of each loss category: central, the shop, an insurer, the transporter, or the supplier?
- F-03** May damaged or expired goods be returned to central or to the supplier for credit, and is there a claim workflow to support that?
- F-04** What approval is required to write off stock, and does the requirement vary by value?
- F-05** Must written-off goods be physically verified and witnessed before disposal, and should the system record who witnessed it?
Why we ask: Write-off is the most common vector for stock fraud in retail, because it makes missing goods legitimate on paper. Dual approval and a witness record are standard controls and are far cheaper to include from the outset.
- F-06** Should stock approaching expiry trigger alerts, and at what lead time by product category?
- F-07** Is markdown selling of near-expiry goods permitted, and should a markdown be recorded as a distinct event separate from an ordinary discount?
Why we ask: Separating the two lets you measure the true cost of over-ordering, which is otherwise hidden inside general discounting and never surfaces to management.
- F-08** Is a disposal or destruction certificate required for any category, such as food, pharmaceutical or agrochemical goods?
- F-09** Should shrinkage be reported as a percentage measure per shop, and is that figure visible to central or only to the shop itself?
Why we ask: This is a politically sensitive metric between independent societies. It belongs in the visibility matrix in Section M rather than being decided by default.
- F-10** Are there acceptable tolerance levels for loss by category, below which no investigation is required?
- F-11** Should the system distinguish goods that are damaged but still saleable at a reduced price from goods that are unsaleable?

Section G. Pricing

G-01 Does central set the retail selling price, recommend it, or leave it entirely to each shop?

G-02 If central sets or recommends prices, may a shop deviate? Does deviation require approval, or merely a recorded reason?

G-03 Are any goods subject to government-controlled maximum retail prices? If so, must the system enforce the ceiling and retain a history of controlled prices?

Why we ask: Selling above a gazetted maximum price is a legal offence, so this is a hard control rather than a preference. It also requires effective-dated price records so that a past transaction can be shown to have been compliant at the time.

G-04 Are there member prices distinct from non-member prices? If so, are they funded by the shop or by central?

G-05 Are promotions required: time-limited price reductions, multi-buy offers, basket-level discounts, bundles?

G-06 Who funds a promotion, central or the shop, and must the system apportion the cost between them?

Why we ask: Between independent entities, promotional funding is a real settlement item rather than a display detail. If central funds a promotion, the shop must be reimbursed, and that calculation has to come from somewhere.

G-07 May till operators apply manual discounts? Within what limits, and with what authorisation?

G-08 Should price changes be effective-dated and scheduled in advance, so that a shop can load next month's prices ahead of time?

Why we ask: Without scheduling, price changes must be applied manually on the day, which across many shops with poor connectivity is a reliable source of error and of pricing disputes with customers.

G-09 Are prices inclusive or exclusive of tax at the shelf edge, and is that consistent across all shops?

G-10 Is rounding required at the till, and to what denomination?

Section H. Point of sale and offline operation

Offline is a requirement, not an option

You have confirmed connectivity will be provided but that outages and slow links are expected in remote areas. That makes offline capability a requirement rather than an option. These questions establish how far it must extend, which is the main cost variable in this section.

- H-01** What is the longest offline period the till must survive while continuing to trade: an hour, a full trading day, several days?
- Why we ask:* An hour can be handled with a simple queue held in memory. Several days requires a full local database at each till, a locally held product and price catalogue, and a designed conflict-resolution strategy. The difference in cost and complexity is large, so the answer should reflect the worst real site rather than the average one.
- H-02** While offline, which operations must still work: sale, refund, member lookup, price lookup, card payment, receipt printing, stock deduction?
- H-03** Which operations may legitimately be blocked while offline?
- H-04** If a shop and central both change the same record while disconnected, whose version prevails on reconnection?
- Why we ask:* Without an explicit rule, a system silently discards one side's data. The usual answer is that a shop is authoritative for its own sales and stock, and central for the catalogue and prices, but this must be confirmed rather than assumed.
- H-05** Are card, mobile wallet or QR payments accepted? These generally require live connectivity, so what should the till do when the payment network is down?
- H-06** How many till lanes are needed per shop, typically and at the largest site?
- H-07** Which peripherals must be supported: barcode scanner, receipt printer, cash drawer, weighing scale, customer display, card terminal, label printer?
- H-08** Is the till to run on a standard computer, a tablet, or a dedicated POS terminal? Is there existing hardware to be reused?
- H-09** Must the receipt print in the customer's chosen language, and may that be selected per transaction?
- Why we ask:* Trilingual printing on low-cost thermal printers has real constraints, particularly for Sinhala and Tamil rendering, and may determine which printers can be purchased. Better decided before hardware is bought.
- H-10** Are refunds, exchanges and voided transactions permitted at the till, and what authorisation do they require?
- H-11** Is a suspended or parked sale needed, so a cashier can set a transaction aside and serve another customer?
- H-12** What are the peak transactions per hour at the busiest shop, and the total daily transaction volume across the federation?
- Why we ask:* Needed to size the central infrastructure and to design the synchronisation approach. An order-of-magnitude estimate is sufficient at this stage.
- H-13** Must a sale be attributable to the individual cashier who processed it?
- H-14** Is home delivery, telephone ordering or any form of online ordering in scope, now or in a later phase?

Section I. Payments, cash and settlement

- I-01** Which tender types are accepted at the counter: cash, card, mobile wallet, member credit account, voucher, coupon, government subsidy instrument?
- I-02** Do shops extend credit to individual members or customers? If so, must the system manage those personal accounts, limits and repayments?
Why we ask: Member credit books are common in cooperative retail and are frequently the largest receivable a shop carries. If in scope this is a substantial module, so it needs to be identified now rather than assumed to be a small feature.
- I-03** Is till cash management required: opening float, cash drops during the day, blind close, till variance reporting?
Why we ask: A blind close, where the cashier counts without seeing the expected figure, is the standard control against till theft. It is a small design choice with a large effect on the usefulness of variance data.
- I-04** Must the system record banking of daily takings and reconcile it against till totals?
- I-05** How does a shop settle its account with central: bank transfer, cheque, deduction from a rebate, offsetting against returns?
- I-06** Should the system produce statements of account between central and each shop, and at what frequency?
- I-07** Should payments be matched to specific invoices, or applied against the oldest balance first?
Why we ask: This determines whether the receivables ledger can produce a meaningful ageing report and support dispute handling on a specific invoice.
- I-08** Are foreign currency or remittance transactions ever involved?

Section J. Inter-entity billing and accounting

On AR and AP

Every supply from central creates a matched pair of entries: an account receivable (AR) at central, being money owed to it, and an account payable (AP) at the shop, being money it owes. Because these sit in the books of two different legal entities, they must agree, and disagreement between them is the routine source of disputes in federated structures. The scope question in J-01 decides how much of that ledger the system owns.

- J-01** Should the system produce a full set of accounts for each shop, that is general ledger, trial balance, profit and loss and balance sheet? Or only manage the receivable and payable between central and each shop, exporting to whatever books each society already keeps?
- Why we ask:* One of the largest scope decisions in the programme. Full multi-entity accounting roughly doubles the finance scope and brings each society's statutory reporting inside the system. Many federations prefer to leave society bookkeeping where it is and integrate instead.
- J-02** If only AR and AP are in scope, what export format is required, and to which accounting packages do the shops currently post?
- J-03** What is the applicable tax regime: VAT, SVAT, or another? Which items are exempt or zero-rated, and does the treatment differ between central and shops?
- J-04** Is government electronic invoicing or fiscal reporting mandated now, or expected in the near term?
- Why we ask:* If it is coming, invoice numbering, immutability and archival must be designed for it from the start. Retrofitting fiscal compliance into a live retail system is disproportionately expensive.
- J-05** What document numbering rules apply, and must numbering be sequential and gapless within each entity?
- Why we ask:* Gapless sequential numbering is a common statutory requirement and it conflicts with offline operation, since a disconnected till cannot know the next number. If it is required, the design must reserve number ranges per site in advance.
- J-06** What is the retention period for transaction and tax records?
- J-07** Are credit notes and debit notes required, and who may issue them?
- J-08** Does the financial year end differ between central and any of the shops?
- J-09** Who audits each society, and are there audit requirements the system must support, such as read-only auditor access or a defined transaction listing?

Section K. Membership and patronage

Why members matter

You asked how members matter to the system. In a cooperative, members are owners rather than ordinary customers, and the relationship often carries financial consequences: preferential prices, patronage rebates on the year's purchases, share capital, credit facilities, and voting rights. If any of these exist, purchases must be attributed to a member identity at the till, which changes the checkout flow and the year-end close. If none do, this section falls away and the system is simply retail. Both are legitimate answers, but it must be answered explicitly.

K-01 Does the business need to distinguish members from ordinary customers at the point of sale at all?

Why we ask: If the answer is no, questions K-02 to K-08 do not apply and a significant amount of scope disappears.

K-02 Does each shop maintain its own member register, or is there an islandwide member identity shared across the federation?

Why we ask: If a member of one society may shop at another, the system needs a federated identity and a rule for which society the patronage accrues to. If not, the register stays local and much simpler.

K-03 Do members receive preferential prices, and are those funded by the shop or by central?

K-04 Are patronage rebates or dividends paid based on a member's purchases over the year, and must this system calculate them?

Why we ask: This requires every relevant sale to be reliably attributed to a member, which puts an identification step into the checkout flow and slows the queue. The operational cost is real and should be weighed against the benefit.

K-05 Is member share capital, subscription or savings tracked, and is that in scope here or held in a separate system?

K-06 Are there member loans or credit facilities to administer?

K-07 Is any governance function expected in the system, such as membership records for AGM voting eligibility, or is that entirely outside scope?

K-08 What personal data is held about members, and what consent and privacy obligations apply to it?

Why we ask: Member records are personally identifiable information belonging to an independent society. Who may see them across the federation is a Section M question with legal weight.

Section L. Central procurement and supplier management

L-01 Is central's own procurement in scope for this system, or handled elsewhere?

L-02 Should shop requisitions and sales history feed a demand forecast that drives central purchase orders?

***Why we ask:** This is one of the strongest arguments for a central system at all, and worth confirming as an explicit objective. It also constrains Section M, because forecasting requires central to see a meaningful amount of shop-level sales data.*

L-03 Must supplier contracts, agreed prices and volume rebates be managed in the system?

L-04 Is there a tender or quotation process central must follow, with an audit trail?

L-05 Are goods imported, requiring landed cost calculation where duty, freight and clearing charges are apportioned across a shipment?

***Why we ask:** Landed cost materially changes the true cost of goods and therefore the transfer price to shops. If imports are significant, this cannot be approximated.*

L-06 Does central manufacture, pack, repack or brand any goods itself?

***Why we ask:** Repacking bulk into retail packs is a production process that consumes one SKU and produces another, and needs its own transaction type rather than being recorded as an adjustment.*

L-07 Are there supplier performance measures to track, such as fill rate, lead time or quality rejections?

Section M. Data ownership, visibility and governance

What this section is really asking

To clarify the earlier question: this is not about whether to use one central database or a separate database per shop. That is an architecture choice made afterwards, and strict ownership rules can be honoured perfectly well inside a single shared database. The question here is the legal and governance one. Because each society is an independent entity, its transaction data is its own commercial record, and the federation must decide who owns it, who may see it, and what happens to it if a society leaves. The technology can implement whatever is decided, but it cannot make the decision.

- M-01** Who legally owns the transaction data generated at a shop: the shop as an independent entity, central, or both jointly?

Why we ask: This governs what central may do with the data, what happens if a society leaves the federation, and who is answerable to a regulator or data protection authority. It is a constitutional question for the federation and should be settled in writing before the system encodes an answer by default.

- M-02** May central see a shop's sales of goods that central did not supply?

- M-03** May central see a shop's locally sourced inventory and the identity of its local suppliers?

- M-04** May central see a shop's margins, profitability and shrinkage?

- M-05** May one shop see any data belonging to another? If lateral transfers are permitted under C-10, stock availability must be visible at minimum. How much beyond that is acceptable?

- M-06** Should there be an aggregate-only tier, where central sees anonymised or combined figures for planning without seeing shop-identifiable detail?

Why we ask: This is frequently the compromise that makes federation-wide reporting politically acceptable. It is straightforward to build if specified now and awkward to retrofit onto reports already written.

- M-07** If a society leaves the federation, what happens to its historical data: deleted, exported to it, or retained by central?

- M-08** Who administers users at a shop: the shop itself, or central?

Why we ask: Shop-level administration respects autonomy but distributes the security risk across many small organisations with limited IT capability. Central administration is safer but slower and may be resisted.

- M-09** Which data protection or privacy obligations apply to member and customer data, and is there a designated responsible officer?

- M-10** Does any authority, such as the Department of Co-operative Development, have a right of access to the data or require returns in a prescribed format?

- M-11** Is there an agreed data sharing agreement between central and the societies, or does one need to be drafted alongside this project?

Why we ask: In most federations this document does not yet exist, and the system project is what forces it to be written. Identifying that early avoids a legal blocker appearing late.

Section N. Security, roles and internal controls

N-01 What roles are required at a shop, for example cashier, supervisor, stock clerk, manager, accountant, and what may each do?

N-02 What roles are required at central?

N-03 How do staff authenticate at the till: password, PIN, card, fingerprint?

Why we ask: Speed at the counter matters, and a control that is slow will be circumvented by sharing logins, which destroys accountability entirely.

N-04 Is multi-factor authentication required for administrative or financial functions?

N-05 Which actions must be irreversible and recorded on an immutable audit trail? For example price override, manual discount, void, refund, stock write-off, opening the cash drawer without a sale, changing a member record.

Why we ask: These are the standard indicators of retail fraud. Deciding now which events are audited is far cheaper than adding auditing later, and an audit trail that can be edited is worth nothing.

N-06 Should the system produce exception reports for management review, and who is responsible for reviewing them?

Why we ask: An exception report nobody owns is a control that does not exist. The reviewer should be named as part of the requirement.

N-07 Are segregation-of-duties rules required, for example that the person who requests a stock write-off may not approve it?

N-08 What happens to a user account when a member of staff leaves, and who is responsible for closing it?

N-09 Are there physical security considerations at shops, such as unattended tills or shared devices, that the design should account for?

Section O. Reporting and analytics

- O-01** For each stakeholder group, what decisions do they need to make, and what information supports each decision? Please answer for central management, a shop manager, and the federation board.

Why we ask: Asking for a list of reports usually produces a long list of screens nobody opens. Naming the decision first produces a much smaller and far more useful set, and makes it obvious when a requested report supports no decision at all.

- O-02** What reporting must remain available offline or on paper at a shop with no connectivity?

- O-03** Is real-time federation-wide reporting genuinely required, or is end-of-day sufficient?

Why we ask: Real-time reporting across many poorly connected sites raises the cost of the synchronisation design considerably. If daily consolidation is adequate, the architecture becomes markedly simpler and cheaper.

- O-04** Are statutory or regulatory returns required in a prescribed format, and by when?

- O-05** Do users need to export to spreadsheets, and should they be able to build their own reports without developer involvement?

- O-06** Is a mobile dashboard needed for managers, given the connectivity constraints?

- O-07** What history must be retained and reportable, and over how many years should trends be comparable?

Section P. Language, localisation and accessibility

Confirmed: all three languages

You have confirmed that Sinhala, Tamil and English must all be supported. These questions determine how deep that requirement runs, because there is a large difference in effort between a trilingual customer receipt and a fully trilingual application with trilingual master data maintained indefinitely.

P-01 Must all three languages be available throughout the user interface, or is the staff interface in one language with only customer-facing output trilingual?

Why we ask: Full interface translation is a one-time cost that recurs at every release. Trilingual master data is a permanent operational commitment. These are different decisions and can be answered differently.

P-02 Is language a per-user setting, a per-shop setting, or switchable at any moment?

P-03 Must reports and printed documents be produced in all three languages?

P-04 Must the system search and sort correctly in Sinhala and Tamil, including for product names and member names?

Why we ask: Correct collation and search behaviour in these scripts is a real engineering task, and it is frequently discovered late when users find they cannot locate a record they know exists.

P-05 What fonts and rendering are acceptable on receipt printers and low-cost shop hardware for Sinhala and Tamil?

P-06 Is data entry required in all three scripts, and will keyboards and input methods be provided at shops?

P-07 Which language is authoritative where a translation is missing, disputed, or where a legal document is concerned?

P-08 Are there literacy or accessibility considerations among shop staff or members that should shape the interface design, for example icon-led screens or larger text?

P-09 Are dates, numbers and currency to be formatted consistently across all three languages?

Section Q. Infrastructure, hosting and service levels

Q-01 Is hosting to be cloud-based, on premises at central, or hybrid? Is there any legal or policy requirement that data remain within the country?

Q-02 Who will own and operate the infrastructure after go-live? Is there an in-house team, or is a managed service expected?

Why we ask: The answer changes the design. A system to be run by a small in-house team must be far simpler to operate than one supported by a vendor under contract.

Q-03 What are the recovery time objective and recovery point objective? That is, how long may the system be unavailable, and how much data may be lost, in a disaster?

Why we ask: These two numbers drive the backup and failover design and are a major cost driver. They should be stated by the business rather than assumed by the technical team, because the business bears the consequence.

Q-04 What availability is expected during trading hours, and is there an agreed service level agreement with defined support hours?

Q-05 What growth is expected over five years in number of shops, users, SKUs and transactions?

Q-06 What connectivity will each shop actually have: fixed broadband, mobile data, or both as failover? What bandwidth is realistic at the weakest sites?

Q-07 Is power reliability a problem at shops, and must the till survive a power interruption on battery-backed hardware?

Why we ask: In practice power interruption is at least as disruptive as connectivity loss, and it affects hardware specification and the safe-shutdown design of the local database.

Q-08 Is there existing infrastructure, licensing or a technical standard the system must fit within?

Q-09 What support model is expected: hours of cover, languages of support, first-line responsibility at shop level, and the escalation path?

Q-10 How will software updates be deployed to shops, and can a shop defer an update?

Why we ask: With many independent sites and poor links, update distribution is a genuine operational problem, and allowing deferral means supporting multiple versions simultaneously.

Section R. Existing systems, data migration and integration

Confirmed position

You have confirmed there is no ERP, and that where shops run a POS today it will be replaced by the one provided with this system. That removes integration with those systems from scope, but not the question of what data must be extracted from them before they are retired.

R-01 Which shops have an existing POS system, which product is it, and can data be extracted from it?

R-02 What historical data must be migrated: product master, member register, opening stock balances, outstanding balances, sales history? How far back?

Why we ask: Opening balances are almost always essential. Deep sales history is usually expensive to migrate and rarely used afterwards. It is worth deciding deliberately rather than migrating everything by default.

R-03 Where does master data currently live: spreadsheets, paper ledgers, individual shop systems? Who owns and maintains it today?

R-04 How clean is the existing data, and who will do the cleansing work?

Why we ask: Data cleansing is very often the critical path of a programme like this and is almost always underestimated. It needs a named owner and dedicated time in the plan, because it cannot be done by the software team alone.

R-05 Must the system integrate with banks for payment files or statement reconciliation?

R-06 Must it integrate with any government portal for tax or cooperative reporting?

R-07 Should an API be provided for future integrations, and are any third parties expected to connect?

R-08 Is SMS or email notification required, for example dispatch alerts, account statements or member communications?

R-09 Are there weighing scales, label printers or other devices already in use at shops that must be supported?

Section S. Programme, rollout and change management

S-01 How many cooperative shops are there in total, and how are they distributed geographically?

S-02 How many staff will use the system, and what is their general level of computer familiarity?

S-03 Is a phased first release acceptable, for example central plus a pilot group of shops with requisition and POS only, or must the full scope go live at once?

Why we ask: Phasing is usually the difference between a programme that lands and one that stalls. Agreeing a minimum viable first release early gives everyone something real to react to within months rather than years.

S-04 Who holds decision-making authority for the federation, and how are decisions affecting all societies ratified?

Why we ask: With independent entities, an unclear decision path is the most common cause of delay on projects of this kind. We need to know who can settle the questions in this document and how long ratification takes.

S-05 Are all shops obliged to adopt the system, or is adoption voluntary?

Why we ask: If voluntary, the system must work correctly with only partial coverage, and central reporting will be incomplete by design. That is workable, but it must be a deliberate choice rather than a surprise.

S-06 What is the target go-live date, and are there seasonal periods such as festivals or harvest that must be avoided?

S-07 Who will deliver training, in which languages, and will it be centralised or delivered regionally?

S-08 How is the system funded: central budget, per-shop subscription, transaction fee? Does the funding model affect what any shop is entitled to use?

Why we ask: A per-shop charge changes the adoption dynamic entirely and may require the system to meter usage, which is a requirement in its own right.

S-09 Who will own the system after go-live, in the sense of prioritising changes and arbitrating between competing requests from societies?

S-10 What does success look like twelve months after go-live, stated in measurable terms?

Why we ask: Writing this down before build begins anchors later scope arguments and gives an honest basis for judging the programme.

Section T. Risks and assumptions to confirm

The following are our working assumptions. Each carries a risk if it proves wrong, and each should be either confirmed or corrected. They are listed separately from the questions because they need a yes or no rather than a considered answer.

Ref	Assumption	Risk if incorrect
T-01	The federation can reach a single binding decision on the questions in Section 3 within a defined timeframe.	Design cannot begin. This is the largest schedule risk in the programme.
T-02	Shops will accept some degree of central visibility over their trading data in exchange for supply and system benefits.	If not, central reporting and demand forecasting lose most of their value and the business case weakens.
T-03	Product master data can be assembled into a single trusted catalogue before go-live.	Cataloguing becomes the critical path and delays everything downstream.
T-04	Shop staff can be trained to operate a computerised till, with support available in their language.	Adoption fails at the counter regardless of how good the system is.
T-05	Connectivity, while imperfect, will be present often enough for daily synchronisation at every site.	Some shops effectively operate permanently offline, which changes the design from occasionally-disconnected to genuinely distributed.
T-06	The societies' existing accounting arrangements can continue unchanged, with this system feeding them.	If full accounting must move into the system, finance scope roughly doubles.
T-07	Hardware at shops will be standardised or at least specified centrally.	Supporting arbitrary hardware across many independent sites is a disproportionate ongoing support burden.

Next steps

We suggest the following sequence. First, circulate Section 3 to whoever holds decision-making authority and obtain written answers to those eight questions, since nothing meaningful can be designed until they are settled. Second, assign each remaining section to the party best placed to answer it, using the guidance in Section 1. Third, return the answers against the question references so that gaps are visible.

Where a question cannot be answered because the federation has never had to decide it, that is a useful finding rather than a failure. Those items should be listed and taken to the board as decisions required, because the system will otherwise make the decision implicitly, and it will make it badly.

Once Section A and Section B are answered, the domain model and the central-to-shop transaction flow can be drafted. That is the backbone to which everything else attaches, and it can be reviewed by the federation before any further commitment is made.

End of document. Draft prepared for discussion and subject to revision as answers are received.